

A Confidence-Based Power-Efficient Framework for Sleep Stage Classification on Consumer Wearables

Abstract:

Consumer wearable devices like smartwatches enable real-time tracking of vital signs with various sensors. Accordingly, experts may leverage smart home techniques for treating sleep disorders by targeting specific sleep stages. To facilitate this scenario, this paper focuses on sleep stage classification based on body movement and heart rate signals detected by wearables in real time. Due to their limited battery capacity, it is crucial to balance the trade-off between power efficiency and classification accuracy. To address the problem, inspired by multi-tasking, we propose COPS, an innovative framework that includes a power-efficient shallow classifier for simple cases and a deep classifier for complex instances. COPS introduces an intelligent switch, CESwitch, to determine a confidence score that directs input to either the shallow or deep classifier. By selectively activating the shallow classifier, the overall expected power consumption could be lower. Two strategies of CESwitch, namely Confidence Delegation and Agreement Verification, are proposed and examined. Notably, both COPS and CESwitch can be seamlessly integrated with existing deep sleep stage classifiers. Comprehensive experimental results on two real datasets manifest that COPS outperforms state-of-the-art lightweight and deep sleep stage classifiers by reducing 77.8% computational cost in terms of FLOPs with only 1.9% accuracy drop. Moreover, adapting to an existing deep model saves up to 32.7% in FLOPs compared to its original architecture.